

Final Experiment Report

Iterative Data Selection for Spider Text-to-SQL

Spider 1.0 + CodeT5-small | 20 September 2026

This mini-project studies whether iterative model-aware data selection can improve Text-to-SQL fine-tuning under a fixed 500-example budget. The study begins with static baselines, adapts the IDU and allocation ideas from LEAD to Spider, and evaluates all methods using five fixed seeds and the official Spider exact-match evaluator.

Executive summary

Five-round IDU-only achieved the strongest selected-data result: **17.08% mean official exact match**, compared with 15.64% for Random and 13.82% for static uncertainty. It won against Random in three of five seeds. The paired mean gain was +1.44 percentage points, but the 95% paired interval (-2.61, +5.49) crossed zero; the result is promising rather than statistically conclusive.

Full-data training remained clearly stronger at 23.02%. IDU-only reduced mean end-to-end time by 23.9% relative to Full Data in the local setup, while losing 5.94 percentage points. The result therefore demonstrates a cost-quality trade-off, not full-data parity.

1. Research questions

- Q1. Can a 500-example selected subset match or exceed training on all 1,000 candidate examples?
- Q2. Does iterative model-aware selection outperform Random under the same selected-data and optimizer-step budget?
- Q3. Does the iterative method improve the end-to-end performance-cost trade-off?
- Q4. Which LEAD-inspired components help in this small Text-to-SQL setting, and which components introduce instability or unnecessary cost?

2. Experimental design

Item	Controlled setting
Dataset	Spider 1.0; 1,000-example training candidate pool; full 1,034-example development set
Model	Salesforce CodeT5-small
Selected-data budget	500 examples for every selected-data method
Seeds	11, 42, 73, 101, 202
Primary metric	Official Spider exact match
Selected-data training	Five-round methods: 100 examples per round; cumulative union; 1,500 total optimizer steps
Cost boundary	Selection/scoring + training + generation + official evaluation

Fairness controls

Random and all iterative selected-data methods use the same 500-example budget, model, seeds, development set, and primary metric. The five-round variants also share the same cumulative training schedule and total optimizer-step budget. Full Data uses all 1,000 examples for three epochs; its larger training cost is included in the end-to-end comparison.

3. Adaptation of LEAD to Spider

LEAD uses iterative model feedback, Instance-Level Dynamic Uncertainty (IDU), and coarse-to-fine allocation with a multi-armed bandit. This project implements a transparent, computationally manageable adaptation rather than a faithful inference-free reproduction.

- Static uncertainty ranks examples once by pretrained target loss and does not update utility.
- Five-round IDU-only selects 100 examples per round, trains on the cumulative selected set, measures observed target-loss change, updates the utility of remaining examples, and selects the next global top-utility batch.
- The full adaptation adds loss-quantile difficulty clusters, EXP3 cluster scheduling, and Spider database IDs as task groups.
- Component ablations separately test fixed static ranking, cluster-MAB allocation, and proportional database-group quotas.

Method boundary: observed loss change is used as a practical IDU proxy. It does not reproduce LEAD's full gradient-based, inference-free utility estimator.

4. Main performance and cost results

Method	Official EM	End-to-end	Databases
Full Data	23.02% +/- 1.83 pp	692.8 s	137.0
Random	15.64% +/- 1.13 pp	502.4 s	126.8
Two-round Iterative	13.56% +/- 1.92 pp	510.3 s	124.0
Five-round Full Adaptation	15.22% +/- 2.96 pp	605.0 s	123.2
Five-round IDU-only	17.08% +/- 2.39 pp	527.3 s	119.2
Static Uncertainty	13.82% +/- 2.37 pp	457.5 s	114.0
Cluster-MAB Component	14.24% +/- 1.13 pp	629.8 s	120.2
Database-quota Component	15.52% +/- 2.04 pp	582.3 s	111.0

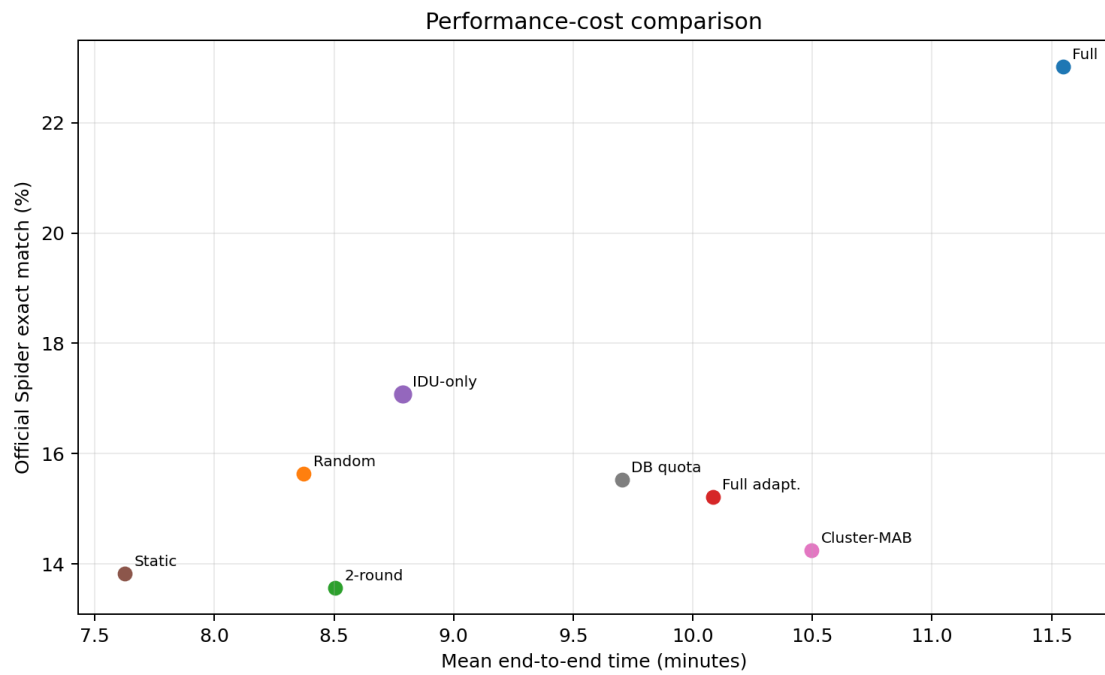


Figure 1. Mean official exact match against measured end-to-end time.

Paired seed analysis

IDU-only minus Random by seed was +1.8, -2.4, +3.5, +5.5, and -1.2 percentage points. The exact sign-flip p-value was 0.4375. With only five seeds, the experiment has low statistical power; the correct conclusion is an observed mean advantage with meaningful seed sensitivity.

Answers to Q1-Q3

Q1: No. The best selected-data method remains 5.94 percentage points below Full Data.

Q2: Partly. IDU-only has the highest selected-data mean and wins three seeds, but the paired interval crosses zero.

Q3: Yes as a trade-off, not as dominance. IDU-only is faster than Full Data but slightly slower than Random while producing a higher observed mean.

5. Difficulty and SQL-component analysis

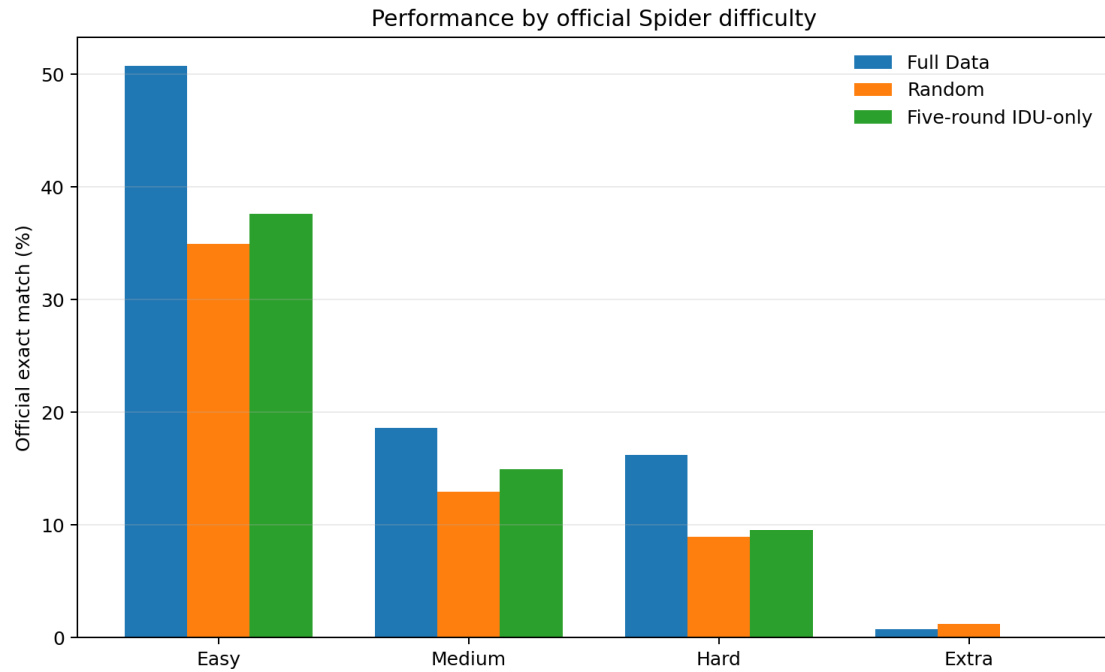


Figure 2. Mean official exact match by Spider difficulty.

Method	Easy	Medium	Hard	Extra	All
Full Data	50.72%	18.62%	16.20%	0.72%	23.02%
Random	34.94%	12.96%	8.96%	1.20%	15.64%
Five-round IDU-only	37.64%	14.94%	9.54%	0.00%	17.08%

IDU-only improves on Random for Easy (+2.70 pp), Medium (+1.98 pp), and Hard (+0.58 pp), but does not solve any Extra examples across the five runs. The benefit is therefore concentrated in less structurally demanding queries.

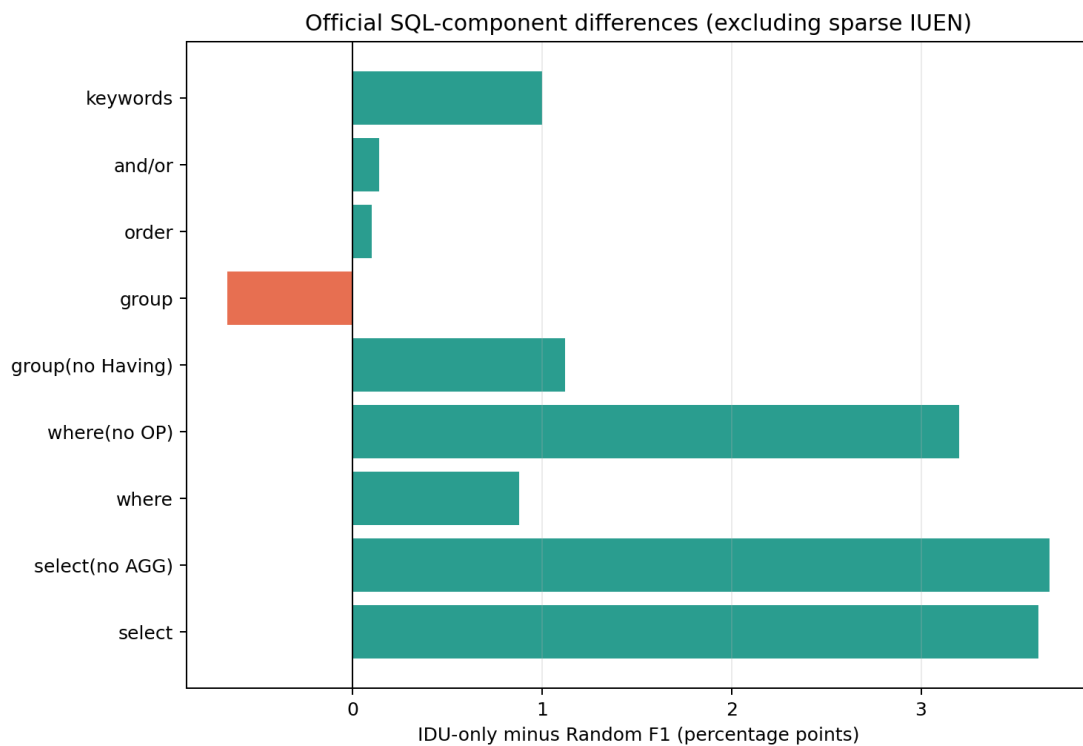


Figure 3. IDU-only minus Random in official partial-match F1; sparse IUEN is excluded from the plot.

The largest stable positive differences are SELECT without aggregation (+3.68 pp), SELECT (+3.62 pp), and WHERE without operator (+3.20 pp). GROUP is slightly lower (-0.66 pp). The official IUEN component is sparse and unstable when a run produces no matching set operation, so its large numerical difference is not interpreted independently.

6. What the ablations show

Comparison	Mean difference	Interpretation
IDU-only vs Static	+3.26 pp	Online utility updating adds value.
IDU-only vs Full Adaptation	+1.86 pp	All allocation components together do not help.
IDU-only vs Cluster-MAB	+2.84 pp	Current clustering and bandit allocation are too coarse or unstable.
IDU-only vs Database quota	+1.56 pp	Proportional quotas reduce useful flexibility.

The central positive result is not that the complete LEAD-style adaptation wins. Instead, the simplest iterative component - updating sample utility from model feedback - is the strongest variant. Adding cluster scheduling or database quotas increases cost without improving the mean.

Selection behaviour

IDU-only and Random have a mean selected-set Jaccard overlap of 0.329. IDU-only covers 119.2 databases on average, versus 126.8 for Random. Its observed advantage does not come from maximising database count; it comes from selecting a different, dynamically updated subset.

7. Manually verified qualitative cases

IDU-only advantage: grouping and HAVING

For 'List all document ids with at least two paragraphs', IDU-only generated the correct Paragraphs + GROUP BY + HAVING structure in four of five seeds. Random failed in all five seeds, often using the Documents table or replacing HAVING with ORDER BY and LIMIT.

IDU-only advantage: schema linking and comparison

For countries becoming independent after 1950, IDU-only produced the direct country/IndepYear > 1950 query in four seeds. Random frequently selected the city table, equality, or the wrong column.

Random advantage: negation

For nationality not equal to Russia, Random preserved the inequality in four seeds. IDU-only failed in all seeds by reversing the condition or generating unnecessary nested NOT IN queries.

Random advantage: literal preservation

For the episode literal 'A Love of a Lifetime', Random preserved the full literal in four seeds. IDU-only repeatedly shortened the literal or selected an invalid output column.

Formatting-only mismatches, such as quotation style or identifier capitalisation, were excluded from the substantive case analysis. These examples explain behaviour but do not replace the official aggregate metric.

8. Limitations

- Five seeds expose instability but are insufficient for a strong statistical superiority claim.
- CodeT5-small has limited capacity and all methods remain weak on Extra queries.
- The observed-loss-change utility is a practical IDU proxy, not LEAD's full gradient-based inference-free estimator.
- MAB and quota variants may be disadvantaged by the small candidate pool, five rounds, and coarse two-cluster grouping.
- End-to-end timing is measured on one Apple MPS environment and should not be treated as hardware-independent throughput.
- Exact match is strict and does not directly measure execution equivalence; auxiliary normalized-string diagnostics are more formatting-sensitive.

9. Conclusion

Under a fixed 500-example budget, five-round IDU-only produced the best selected-data mean at 17.08%, 1.44 percentage points above Random. The study provides preliminary evidence that updating sample utility as the model changes is more useful than a fixed uncertainty ranking. However, the advantage is seed-sensitive and does not close the gap to Full Data. The complete cluster-MAB and database-allocation variants do not improve the result in this setting.

The most defensible outcome is therefore a controlled negative-and-positive finding: iterative utility updating is promising, while the additional allocation mechanisms require better cluster definitions, reward design, or larger-scale validation before they can be claimed to help Text-to-SQL.

10. Recommended next work

1. Manually classify a larger sample of official errors, focusing on negation, literals, joins, GROUP/HAVING, and set operations.
2. If additional compute is available, repeat IDU-only with more seeds or one larger model before adding further selector complexity.
3. For a closer LEAD reproduction, replace the observed-loss-change proxy with the paper's gradient-based IDU estimator and redesign clusters around Text-to-SQL structure rather than only loss quantiles.

References

- [1] T. Yu et al. Spider: A Large-Scale Human-Labeled Dataset for Complex and Cross-Domain Semantic Parsing and Text-to-SQL Task. EMNLP, 2018.
- [2] Y. Wang et al. CodeT5: Identifier-aware Unified Pre-trained Encoder-Decoder Models for Code Understanding and Generation. EMNLP, 2021.
- [3] X. Lin et al. LEAD: Iterative Data Selection for Efficient LLM Instruction Tuning. PVLDB 19(3):426-439, 2025. doi:10.14778/3778092.3778103.

Evidence boundary: all numerical results in this report come from completed local experiments. The work is a simplified LEAD-style adaptation and does not claim faithful reproduction of the paper's full estimator or large-scale results.